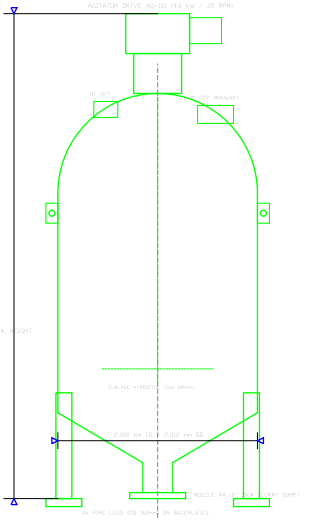
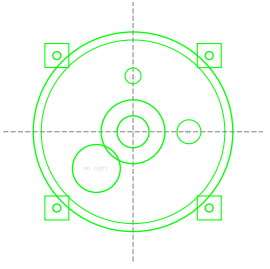
MARK	SIZE	RATING	FACING	SERVICE / DESCRIPTION
N1	6" NB	150# ANSI	RF, WNRF	Raw Dal Grain Infeed Chute
N2	2" NB	150# ANSI	RF, WNRF	Process Soft Water Inflow
N3	1" NB	150# ANSI	RF, WNRF	Chemical Reagent Dosing Inflow
N4	6" NB	150# ANSI	RF, WNRF	Soaked Dal Slurry Dump to Screen
N5	2" NB	150# ANSI	RF, WNRF	CIP Spray Ball Supply Header
N6	3" NB	150# ANSI	RF, WNRF	Atmospheric Vent / Pressure Relief
M1	20" NB	150# ANSI	RF, Davit	Inspection & Maintenance Manway

MECHANICAL DESIGN DATA & FABRICATION SPECIFICATION

1. DESIGN CODE: ASME Boiler & Pressure Vessel Code Sec VIII Div 1 (References)
2. WORKING CAPACITY: 7.50 m3 | GROSS INTERNAL VOLUME: 8.49 m3
3. DESIGN PRESSURE: Full Atmospheric (FV / 0.50 bar g Hydrostatic Liquid Head)
4. DESIGN TEMPERATURE: 65.0 deg C | OPERATING TEMPERATURE: 30.0 deg C +/- 1 deg C
5. SHELL METALLURGY: AISI 316L, 10.0 mm nominal thk, 2B sanitary finish)
6. HEAD METALLURGY: AISI 316L, 10.0 mm thk, 2:1 Ellipsoidal seamless formed)
7. CONICAL HOPPER: AISI 316L, 10.0 mm thk, 60 deg included angle with smooth knuckle)
8. INTERNAL SURFACE FINISH: Ra +/- 0.8 um (120 Grit electro-polished, weld ground flush)
9. EXTERNAL FINISH: Bead-blast, passivated and Scotch-Brite uniform satin finish.
10. RADIOGRAPHY & NDT: Spot Radiography per ASME UW-52 (Joint Efficiency E = 0.85).
11. HYDROSTATIC TEST: Shop Hydro-tested at 1.30 x Design = 0.65 bar g For 60 min.
12. AGITATOR: SEW-Eurodrive Helical-bevel gearhead motor, 4.0 kW, 35 RPM output.
13. IMPELLER: 3-Blade high-efficiency hydrofoil, Ø1x 800 mm, AISI 316L solid cast)
14. CIP SYSTEM: 2x 360 deg Rotary spray balls (316L) covering 100% internal surface.
15. VESSEL WEIGHTS: Tare Empty Weight (1,850 kg) Full water-filled Weight: 9,350 kg.
16. FOUNDATION LOADS: 4x Support legs, vertical load 23.4 kN/leg, anchor bolts 4x M20 316L.
17. CERTIFICATION: Manufacturer's Data Report Form U-1 / EN 10204 3.1 Mill Test Certificates.



FRONT ELEVATION VIEW (SCALE 1:20)



TOP PLAN VIEW (SCALE 1:20)

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - SOAK VESSEL TK-101
PROCESS MECHANICAL | STAGE: CONCEPTUAL FEED

DWG NO: DWG-FIN-IST-007	REV: V4.0	SHEET: 7 OF 14
SCALE: AS SHOWN	DATE: 2026-03-15	STATUS: PRELIMINARY